Merits of Mode:

- It is comparatively easy to understand.
- It can be found graphically.
- It is easy to locate in some cases by inspection.
- It is not affected by extreme values.
- It is the simplest descriptive measure of average.

Demerits of Mode:

- It is not suitable for further mathematical treatment.
- It is an unstable measure as it is affected more by sampling fluctuations.
- Mode for the series with unequal class intervals cannot be calculated.
- In a bimodal distribution, there are two modal classes and it is difficult to determine the values of the mode.

5.4 Empirical Relationship among mean, median and mode

A frequency distribution in which the values of arithmetic mean, median and mode coincide is known of symmetrical distribution, when the values of mean, median and mode are not equal the distribution is known as asymmetrical or skewed. In moderately skewed asymmetrical distributions a very important relationship exists among arithmetic mean, median and mode.

Karl Pearson has expressed this relationship as follows

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Arithmetic Mean}$$

Example 5.28

In a moderately asymmetrical frequency distribution, the values of median and arithmetic mean are 72 and 78 respectively; estimate the value of the mode.

Solution:

The value of the mode is estimated by applying the following formula:

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean} = 3\,(72) - 2\,(78)$$

$$= 216 - 156 = 60$$

$$\text{Mode} = 60$$

Ch5_11th Stats.indd 141 2/3/2019 11:01:04 AM